MIL-R-3043B

5 January 1972

SUPPERSEDING
MIL-R-3043A
28 September 1965

MILITARY SPECIFICATION

RESIN COATING, PERMANENT, FOR ENGINE COMPONENTS AND METAL PARTS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of permanent resin coating for surfaces of engine components and metal parts. The specified coating conforms to air pollution regulation requirements.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

P-D-680	Dry Cleaning Solvent
QQ-M-44	Magnesium Alloy Plate and Sheet (AZ31B)
QQ-S-698	Steel, Sheet and Strip, Low Carbon
TT-N-97	Naphtha; Aromatic
TT-P-143	Paint, Varnish, Lacquer, and Related Materials, Packaging, Packing and Marking of
TT-S-735	Standard Test Fluids, Hydrocarbon

Military

MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on Remover, Paint and Varnish (Silicate Type)
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-C-25107	Carbon Removing Compound, Orthodichlorobenzene, for Engine Parts

FSC 8030

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STANDARDS

Federal

Fed Test Method Std No. 141	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Color Index, Second Edition, Vol 3, Part 4

(The AATCC Technical Manual may be consulted in libraries, and is obtainable from the AATCC National Headquarters, P. O. Box 12215, Research Triangle Park, North Carolina 27709. Reprints of Part 4 are obtainable from the same source.)

3. REQUIREMENTS

3.1 Material. The base material shall be a thermosetting resin, free from drying and non-drying oils and cellulose.

3.2 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate department medical service who will act as an advisor to the procuring activity.

3.3 Color. The material shall be dyed blue by the addition of not less than 0.09 percent nor more than 0.11 percent of Victoria Blue B base spirit soluble dye, COLOR INDEX designation C. I. Solvent Blue 4, C. I. Number 44045B (COLOR INDEX Second Edition, Vol 3, Part 4, page 3377). This percentage shall be based on the finished material. Other oil soluble dyes which give the same color intensity at the same con-

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centration as Victoria Blue B base will be acceptable at the discretion of the procuring activity. The coating shall exhibit a distinct green coloration after the baking period and shall be sufficiently transparent so that the metal surface underneath the film will be clearly discernible.

3.4 Solids content. The solids content of the coating as furnished shall be not less than 25 percent nor more than 30 percent by weight when tested in accordance with 4.5.1.

3.5 Viscosity. The viscosity of the coating as furnished shall not exceed 0.4 poise when tested in accordance with 4.5.1.1.

3.6 Volatile content. The volatile content of the coating shall consist of a non-photochemically reactive solvent blend. A non-photochemically reactive solvent is any solvent with an aggregate of less than 20 percent of its total volume composed of the chemical compounds classified below or which does not exceed any of the following individual percentage composition limitations, referred to the total volume of solvent:

a. A combination of hydrocarbons, alcohols, aldehydes, esters, ethers or ketones having an olefinic or cycloolefinic type of unsaturation: 5 percent.

b. A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene: 8 percent.

c. A combination of ethylbenzene, ketones having branched hydrocarbon structures, trichloroethylene or toluene: 20 percent.

3.7 Performance requirements.

3.7.1 Application. The material shall be capable of being applied to metallic surfaces by dip or spray at room temperatures, 18° C to 29° C (65° F to 85° F).

3.7.2 Drying time.

3.7.2.1 Air dry. The coating shall dry at room temperatures, 18° C to 29° C (65° F to 85° F) in not more than 30 minutes when tested in accordance with 4.5.10.1.

3.7.2.2 Oil bake. The coating air dried at room temperature 16 to 24 hours shall bake hard and tough and be free of bubbles, pinholes or other film irregularities when immersed in hot lubricating oil as described in 4.5.10.2.

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3.7.2.3 Oven bake. The coating shall bake hard and tough and be free of bubbles, pinholes or other film irregularities after baking in accordance with 4.5.10.. Test panels prepared for test of spray application, especially, shall be examined for these defects. A slight orange peel effect on these panels is acceptable.

3.7.3 Flexibility. The coating on a finished panel, when prepared in accordance with 4.4 and tested in accordance with 4.5.2, shall show satisfactory adherage at the bend when rapidly bent over a 1/4-inch rod. It shall show no tendency to chip off in large pieces or peel when scratched with a sharp knife. Slight cracking at the bend will be permitted.

3.7.3.1 Abrasive flexibility. The film on a specimen shall show no chipping or peeling when prepared in accordance with 4.4 and tested in accordance with 4.5.11. Wearing away of the film by abrasion will be permitted.

3.8 Environmental resistance.

3.8.1 Humidity. Test panels prepared in accordance with 4.4 and tested in accordance with 4.5.3 shall show no evidence of corrosion or spread of corrosion beyond that in the scratches cut in the finish, and the film shall show no softening, peeling, blistering, chipping, or loss of adhesion.

3.8.2 Salt spray. Test panels prepared in accordance with 4.4 and tested in accordance with 4.5.4 shall show no evidence of corrosion or spread of corrosion beyond that in the scratches cut in the finish and the film shall show no evidence of softening, peeling, blistering, chipping, or loss of adhesion.

3.8.3 Aromatic hydrocarbon. Test panels prepared in accordance with 4.4 and tested in accordance with 4.5.5 shall show no evidence of film failure. The change in weight shall not exceed 0.002 gm.

3.8.4 Salt water-hydrocarbon. Test panels prepared in accordance with 4.4 and tested in accordance with 4.5.6 shall show no evidence of softening, blistering, leaching, or loss of adhesion and there shall be no evidence of corrosion on the metal.

3.8.5 High humidity - cold cycle. Test panels prepared in accordance with 4.4 and tested in accordance with 4.5.7 shall show no evidence of corrosion beyond that in the scratches cut in the finish, and the film shall show no softening, peeling, blistering, chipping, or loss of adhesion.

3.8.6 Hot oil. Test panels prepared in accordance with 4.4 and tested in accordance with 4.5.8 shall show no evidence of film failure and the change in weight of the specified test panel shall not exceed 0.005 gram. Darkening of the film shall not be cause for rejection.

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3.8.7 Carbon remover. Test panels prepared in accordance with 4.4 and tested in accordance with 4.5.9 shall show no flaking or peeling.

3.9 Removability. Films shall be completely removed from the test panel within one hour when tested in accordance with 4.5.12.

3.10 Workmanship. The workmanship shall be in accordance with high grade commercial practice covering this type of material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certification of conformance to air pollution regulation requirements. In addition to the inspection requirements specified herein, the supplier shall provide a certified statement that the solvents used in the manufacture of the coating and the solvent mixture content of the finished coating meets the requirements of 3.6.

4.2 Classification of tests. The inspections and tests of the coating are classified as quality conformance tests.

4.3 Sampling for inspection and testing.

4.3.1 Lot. For the purpose of sampling, a lot shall consist of all material offered at one time for acceptance by the Government.

4.3.2 Sampling. A random sample of containers of coating shall be taken from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) of 4.0 percent defective to verify compliance to the requirements of this specification.

4.4 Preparation of test panels. Metal test panels of steel and magnesium shall be prepared for all tests as noted below except that for spray application tests and for flexibility tests only steel shall be used. The 1¼ by 4-inch and 2 by 4-inch panels shall be 0.063 to 0.125 inch thick, the 3 by 6-inch and the 6 by 12-inch panels (for spray tests) shall be 0.014 to 0.030 inch thick, steel shall conform to QQ-S-698; magnesium shall conform to QQ-M-44. All edges shall be rounded and a ¼-inch hole shall be drilled on the center line near one end for all panels except the 2 by 4-inch panels. Remove all burrs, sharp edges and corners, including those at the

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edges of holes. The 1¼ by 4-inch and the 2 by 4-inch panels shall be surface ground to a roughness of 10 to 16 micro-inches root mean square. After grinding, panels shall be immediately slushed in a water displacing preservative compound to remove coolant used in grinding. The 3 by 6-inch and 4 by 12-inch panels shall be smoothly finished with a No. 320 grit medium. In addition, magnesium panels shall be given a type III treatment in accordance with MIL-M-3171. Unless the steel panels are to be used immediately they shall be preserved in an atmosphere having a relative humidity not greater than 20 percent or in a container of fresh preservative compound. Magnesium panels, however, should not be preserved in oil. Immediately prior to use the panels shall be cleaned by successive washings in solvent conforming to P-D-680, drained thoroughly, followed by slushing in boiling 95 percent methanol, and finally in boiling absolute methanol. The 6 by 12-inch panels should be cleaned thoroughly with clean solvent-soaked lintless rags. Care must be taken during cleaning and preparation so that the surface is not contaminated by fingerprints. The panels shall be handled by tongs or hooks during the cleaning and dipping operations. The size of panels for each test shall be as follows:

Test	Panel Size
Spray application	6 by 12-inch
Dip application	optional
Drying time	optional
Baking in oil	1¼ by 4-inch
Oven baking	6 by 12-inch
Flexibility	3 by 6-inch
Abrasive flexibility	3 by 6-inch
Humidity	2 by 4-inch
Salt spray	3 by 6-inch
Aromatic hydrocarbon	1¼ by 4-inch
Salt water hydrocarbon	3 by 4-inch
High humidity cold cycle	2 by 4-inch
Hot oil	1¼ by 4-inch
Removability	optional
Sandblast	optional
Carbon remover test	3 by 6-inch

4.4.1 Application of coating. All test panels except the spray application test panels shall be prepared by dip coating in an atmosphere at a relative humidity not above 65 percent and a temperature of 18° to 29° C (65° to 85° F). The clean test panels shall be withdrawn vertically at a rate so as to produce the dry film coating thicknesses of 0.5 to 0.7 mils for all tests except flexibility which shall be maintained with 0.2 to 0.3 mils. These thicknesses may require successive dipping at withdrawal rates of 4 to 40 inches per minute. Accurate film thicknesses can be produced after a little experimentation. The 6 by 12-inch panels shall be coated by spraying to a film thickness of 0.5 mil ± 0.2 mil. These panels shall then be allowed to

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air dry for not less than one hour at the above ambient conditions. All panels except the "Baking in Oil" and "Oven baking" test panels shall be baked prior to test at 163° C ± 3° C (325° F ± 5° F) for 30 ± 5 minutes.

4.5 Test methods.

4.5.1 Solids content. The test shall be performed in accordance with FTMS No. 141, Method 4041.

4.5.1.1 Viscosity. The test shall be performed in accordance with FTMS No. 141, Method 4287.

4.5.2 Flexibility. The test shall be performed in accordance with FTMS No. 141, Method 6221.

4.5.3 Humidity resistance. The test shall be performed in accordance with FTMS No. 141, Method 6201. One face of the oven baked test panels and one face of the oil baked test panels shall be cross scratched (X) with a sharp instrument so that the underlying metal is exposed at least 2 inches along the scratch and suspended in a humidity cabinet in an atmosphere of 100 percent relative humidity at 49° C ± 1° C (120° F ± 2° F) for 10 days. Upon removal, the panels shall be wiped dry with lintless cloths and examined for corrosion and failure of the film.

4.5.4 Salt spray resistance. The test shall be performed in accordance with FTMS No. 141, Method 6061. The test panels shall be inclined 15 degrees from the vertical, cross hatched (X) on one face with a sharp instrument so that the underlying metal is exposed at least 2 inches along the scratch and shall be subjected to a 5 percent salt spray test for 10 days.

4.5.5 Aromatic hydrocarbon resistance. The test shall be performed in accordance with FTMS No. 141, Method 6011. The test panels shall be weighed and then immersed in a flask containing hydrocarbon test fluid conforming to Type III of TT-S-735. The flask shall be connected to a reflux and the contents of the flask heated to 65.6° C (150° F) for three periods of 8 hours each for a total of 24 hours, without removal of the panels. At the completion of the test the panels shall be removed, air dried 1 hour and weighed.

4.5.6 Salt water-hydrocarbon resistance. The test shall be performed in accordance with FTMS No. 141, Method 6011. The test panels shall be immersed vertically for seven days in a covered glass vessel containing a two layer liquid, consisting of a 3 percent aqueous sodium chloride solution and hydrocarbon test fluid conforming to Type III of TT-S-735, so that 2 inches of the panel is exposed to the salt mixture, 2 inches of the panel exposed to the aromatic hydrocarbon test fluid and the balance of the panel exposed to the air-vapor mixture. The temperature during the test shall be maintained at 35° C to 38° C (95° F to 100° F). Immediately upon removal from the liquid, the panel shall be examined.

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4.5.7 High humidity-cold cycle. The test shall be performed in accordance with FTMS No. 141, Method 6201. The test panels cross scratched (X) on one face with a sharp instrument shall be subjected to a cycle of 8 hours in an atmosphere of 100 percent relative humidity at 49° C ± 1° C (120° ± 2°F) followed by 16 hours at -54° C (-65° F) for 7 days. The test cycle may be interrupted to conform to work periods; however, this break should occur only after the cold period. Upon completion of this test the panels shall be examined

4.5.8 Hot oil resistance. The test shall be performed in accordance with FTMS No. 141, Method 6011. Weighed test panels shall be immersed in lubricating oil conforming to MIL-L-7808, and heated to 204° C (400° F) in less than two hours and maintained at that temperature for three periods of 6 hours each. The panels shall remain immersed in the oil overnight between heatings. At the completion of the tests the panels shall be removed, the oil cleaned off with naphtha conforming to TT-N-97, air dried for 1 hour and weighed.

4.5.9 Carbon remover resistance. The test shall be performed in accordance with FTMS No. 141, Method 6011. The test panels shall be immersed in material conforming to MIL-C-25107 maintained at 57.2° to 62.8° C (135° to 145° F) for three successive periods of 6 hours each. At the end of these immersions, the panels shall be observed for defects.

4.5.10 Drying time.

4.5.10.1 Air dry. Coated test panels air dried at room temperature 18° C to 29° C (65° F to 85° F) for 30 minutes shall be dry to touch with no transfer of coating.

4.5.10.2 Oil bake. Coated test panels air dried for 16 to 24 hours shall be immersed to a depth of approximately two-thirds of their length in hot lubricating oil conforming to MIL-L-7808 at 177° C ± 6° C (350° F ± 10° F) for 15 minutes. The panels shall then be removed, rinsed clean with naphtha, and the finish examined for film irregularities. Darkening of the finish or difference of color in the immersed and unimmersed portion shall not be cause for rejection.

4.5.10.3 Oven baking. The test panels, coated by spraying and air dried for not less than 1 hour, shall be placed in an oven and baked for 30 ± 5 minutes at 163° C ± 3° C (325° ± 5° F).

4.5.11 Abrasive flexibility. The coating on a test panel shall be subjected to a sandblast provided by any suitable means. The sandblast must be such that it abrades the coating. Operating air pressure should be 40 psig.

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4.5.12 Removability. The coated panels shall be immersed in a remover conforming to MIL-R-7751 diluted as follows: 8 ounces stripping compound per gallon of water. The diluted solutions shall be maintained at 93° ± 3° C (200° F ± 5° F). A soft bristle brush may be used in the stripping operation in order to facilitate the removal of the loosened finish. From the stripping bath the panels shall immediately be transferred to a bath of boiling distilled water containing 0.02 to 0.05 percent NaoH and thoroughly rinsed. Immediately after this rinse the panels shall be rinsed in clean solvent conforming to P-D-680 and finally in 95 percent ethyl or methyl alcohol and diethyl ether. The time required to completely strip the panels shall be noted. In addition, the stripped panels shall be examined for evidence of corrosion.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The resin coating shall be packaged in containers as required by TT-P-143.

5.1.2 Level C. Packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The resin coating shall be packed in overseas type shipping containers according to the requirements of TT-P-143.

5.2.2 Level C. The resin coating shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Interior and exterior containers shall be marked in accordance with the requirements of MIL-STD-129. In addition, individual containers shall be marked, either by stencil or durably attached labels, with mixing, thinning, application, and curing instructions, as applicable, date packaged and lot number.

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6. NOTES

6.1 Intended use. Material covered by this specification is intended for use as a permanent corrosion preventive, oil-resistant coating for metallic non-bearing surfaces of engine parts and other components. However, this material may be applied to gear teeth and such other bearing surfaces where wearing off of the film will cause no detrimental effects and where tolerances greater than 0.001 inch are encountered.

6.2 Ordering data. Procurement documents should specify:

Title, number, and date of specification
Size of the container
Applicable level of packaging and packing (see 5.1)

6.3 The material shall be purchased by volume, the unit being a U. S. gallon (231 cubic inches) at 25° C (77° F).

6.4 Storage life. The storage life of materials conforming to this specification is 24 months when stored at temperatures of 80° F or less.

6.5 The baking temperatures given in this specification are intended to control test procedures only. Due to the influence of size and shape of coated articles, baking facilities and other production factors, the time and temperature relationship for production operations should be adjusted to produce baked films equal in all properties to those obtained on the test panels under the specified test conditions.

6.6 Inasmuch as this material is an electrical insulator, parts to be tested by magnetic particle inspection may require stripping at the terminals depending on the method of magnetic inspection employed. The presence of this coating tends to reduce the intensity of indications or otherwise to interfere with the detection of the physical defects normally exposed by magnetic particle inspection of uncoated parts.

Custodians:
Army - MR
Navy - AS
Air Force - 84

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B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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